

SIDDHI DABHOLKAR

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SUMMARY

Software Engineer with 2+ years of experience shipping production features across enterprise and startup environments, building full-stack web applications and backend services from design to deployment. Actively building RAG pipelines, hybrid vector search, and agentic workflows - from retrieval design through evaluation and deployment.

EDUCATION

Master of Science in Computer Science | *University of Illinois Chicago*
Bachelor of Engineering in Computer Engineering | *University of Mumbai*

Aug 2024 - May 2026
Aug 2018 - May 2022

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C#, Shell Scripting
Backend: Spring Boot, Hibernate, JSF/J2EE, Node.js, ASP.NET, PHP, ActiveMQ
Frontend: React, HTML, Tailwind CSS, Bootstrap, RESTful APIs
Databases: Microsoft SQL Server, PostgreSQL, MongoDB, SQLite, Redis
DevOps: Linux, Docker, AWS (EC2, S3), GitHub Actions, Vercel
QA & Tools: WalkMe DAP, JUnit Testing, Jira, Power BI

PROFESSIONAL EXPERIENCE

Software Engineer Intern | *Bectran Inc., Illinois*

May 2025 - Aug 2025

- Contributed to JDK 17/Tomcat 10 migration of legacy Spring Boot applications, leveraging Amazon Q for compatibility analysis, refactoring code, and resolving dependency conflicts, improving system performance by 25%.
- Standardized system report exports by modifying controller logic to generate CSV outputs and deliver files via SFTP, eliminating 30% of downstream compatibility issues and accelerating reporting workflows.
- Fixed invoice generation and dynamic email template defects across microservices and ActiveMQ message queues, improving accuracy of cron-scheduled customer document generation and reducing manual corrections by 30%.

Software Engineer | *LTIMindtree, Mumbai*

Jun 2022 - Jul 2024

- Built 47+ enterprise workflow automation flows using WalkMe, JavaScript, and jQuery for platforms like Salesforce and Dynamics 365, reducing process completion time by 55% and ServiceNow support tickets by 45%.
- Collaborated with 5 cross-functional teams in an Agile environment to define sprint scope, conduct UAT, and execute releases, translating requirements into production-ready features.
- Employed WalkMe Insights to deliver adoption analytics (funnels, completion rates, drop-offs), pinpoint workflow bottlenecks, and drive changes that improved system stability.
- Developed RESTful APIs using ASP.NET for a marketplace platform with React-based frontend integration, supporting product catalog management, order processing, and vendor registration, reducing third-party vendor dependency by 50%.
- Designed and managed SQL Server database schemas, entity relationships, and transactional workflows using Entity Framework to support product, vendor, and order management modules.

Software Developer Intern | *Fouses IT Solutions, Patna*

Jun 2021 - Aug 2021

- Developed a full-stack dynamic news portal using PHP and MySQL with role-based access control to publish, manage, and curate content. Integrated Google OAuth 2.0 for secure user authentication.
- Built interactive features such as likes, comments, keyword-based search, and category-level filtering, improving user engagement and content discoverability by 15%.

PROJECTS

CaseQuery - Legal RAG Assistant | *Sentence-Transformers, FAISS, BM25, OpenRouter, Python*

- Built CaseQuery, a legal QA system over CUAD contracts and Illinois CAP case law using hybrid retrieval (FAISS dense embeddings and BM25 sparse) and Llama-2 for source-attributed answer generation.
- Containerized and deployed the Flask application on AWS EC2 using Docker and GitHub Actions, integrating an evaluation pipeline comparing FAISS vs. BM25 (Recall@K, MRR), and RAG vs. zero-shot generation, achieving a BERTScore of 0.857

PaperPilot AI | *LangGraph, Pinecone, FastAPI, React, DeepSeek*

- Built a multi-agent RAG system over 800+ NLP/ML papers using LangGraph agents for query decomposition, retrieval, critique, reformulation, synthesis, and verification, achieving 0.765 average citation recall and 0.84 answer relevancy.
- Optimized semantic retrieval accuracy by implementing Reciprocal Rank Fusion across BM25 and dense Pinecone vectors (bge-large), fueled by a custom GROBID PDF ingestion pipeline that extracts section-aware chunks and Semantic Scholar metadata.
- Built a FastAPI-React interface that streams real-time synthesized answers and renders inline arXiv citations linking directly to cited research papers.

Inn-Sight Web Application | *MongoDB, Express, React.js, Node.js, RESTful API, Axios*

- Developed a full-stack (MERN) accommodation booking platform with JWT authentication, protected user flows, and React Router navigation across listing browsing and booking management.
- Designed 15+ REST APIs in Node.js/Express for users, listings, and bookings; integrated Mongoose validation, Multer + AWS S3 for image uploads, and Axios for API consumption on the frontend.